

WEEK 6 CAPSTONE PROJECT

Integrative Data Science Pipeline for Wine Quality Analysis

Data Acquisition • Cleaning • EDA • Regression • Clustering • Evaluation

Python | Pandas | Matplotlib | Seaborn | Scikit-learn

Project item	Details
Dataset	UCI Wine Quality — red vinho verde samples
Source	UCI Machine Learning Repository
Rows used after duplicate removal	1,359
Predictive task	Regression: predict sensory quality score
Descriptive task	K-Means clustering with k=2
Primary model	Random Forest Regressor
Secondary benchmark	Gradient Boosting Regressor
Primary evaluation	MAE, RMSE, R ² , 5-fold cross-validation

1. Executive Summary

This capstone integrates the main stages of a practical data science workflow using the UCI Wine Quality dataset. The dataset contains physicochemical measurements for red vinho verde wines and a sensory quality score. The UCI repository describes the dataset as suitable for both regression and classification; the red subset contains 1,599 observations, 11 physicochemical input variables, and no missing values. The official record also notes that quality scores are ordered and imbalanced, making the problem a useful example of real-world tabular modeling.

After acquisition, the raw data were checked for missing values, duplicates, data types, implausible values, and feature scale differences. The source file contained 1,599 rows and 12 columns. 240 duplicate rows were identified and removed, leaving 1,359 unique observations. Exploratory analysis showed that quality scores are concentrated around the middle of the scale, while several physicochemical variables have different ranges and moderately strong relationships with quality. The predictive component used a leak-safe Scikit-learn pipeline with standardization and a Random Forest Regressor. A Gradient Boosting model was trained as a benchmark.

On the held-out 20% test set, the Random Forest achieved $MAE=0.491$, $RMSE=0.635$, and $R^2=0.399$. Five-fold cross-validation produced mean $MAE=0.492 \pm 0.013$ and mean $R^2=0.397 \pm 0.033$. The unsupervised component used standardized physicochemical features and K-Means; the highest silhouette score among $k=2-6$ occurred at $k=2$ (silhouette=0.205), producing interpretable profiles that differ mainly in alcohol, acidity, sulphates, and sulfur-dioxide variables.

- Use a single reproducible dataset and preprocessing logic for both supervised and unsupervised analysis.
- Keep transformations inside model pipelines to reduce leakage risk and improve reproducibility.
- Use both predictive metrics and cluster structure metrics rather than relying on one score.
- Interpret model outputs as analytical evidence rather than causal proof; sensory quality is influenced by factors not present in the dataset.

2. Problem Statement and Project Goals

Problem statement: Given measurable physicochemical properties of red wine, can we predict its sensory quality score accurately enough to support screening and analytical decision-making? In parallel, can the same feature space reveal natural groups of chemically similar wines without using the quality score?

- Build a supervised regression model that predicts the numeric quality score.
- Use cross-validation and a held-out test set to estimate generalization performance.
- Identify influential predictors using permutation importance.
- Perform K-Means clustering without using the target label, then profile and visualize the resulting groups.
- Connect predictive and descriptive findings into practical recommendations and limitations.

3. Methodological Approach

1. Data acquisition from the official UCI Machine Learning Repository.
2. Data quality assessment: shape, types, missing values, duplicates, ranges, and outlier flags.
3. Exploratory analysis: distributions, correlations, and feature-versus-quality comparisons.
4. Supervised learning: 80/20 stratified split, Random Forest regression, Gradient Boosting benchmark, 5-fold cross-validation.
5. Unsupervised learning: standardization, K-Means model selection with silhouette score, PCA visualization, and cluster profiling.
6. Evaluation and synthesis: compare performance, examine errors, identify practical implications, and document limitations.

4. Data Collection and Acquisition

The project uses the red-wine portion of the UCI Wine Quality dataset. The official UCI record states that the dataset concerns vinho verde samples from Portugal and provides 11 physicochemical input variables plus a sensory quality output score. The dataset is licensed under CC BY 4.0. The data were downloaded as a CSV file and read using Pandas.

```
import pandas as pd
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/winequality-red.csv"
```

```
df = pd.read_csv(url, sep=";")
print(df.shape)
print(df.head())
```

Source: UCI Machine Learning Repository, Wine Quality dataset (Cortez et al., 2009).

5. Data Cleaning and Preprocessing

5.1 Structural checks

- Raw shape: 1,599 rows x 12 columns.
- Missing values: 0 total missing cells were found.
- Duplicate rows: 240 exact duplicate rows were identified.
- All analytical variables were numeric after CSV parsing.
- Quality is treated as the target for supervised learning and is excluded from clustering features.

```
# Quality checks
print(df.info())
print(df.isna().sum())
print("duplicates:", df.duplicated().sum())
print(df.describe().T)
```

5.2 Duplicate handling

240 exact duplicates were removed before modeling. The rationale is to prevent repeated records from receiving disproportionate weight during training and to reduce the risk of putting identical observations into both training and test folds.

```
df_clean = df.drop_duplicates().reset_index(drop=True)
```

5.3 Range and outlier review

No row was deleted solely because it was statistically unusual. Instead, IQR-based counts were used as diagnostic flags. This conservative choice is appropriate because extreme chemical values can be valid observations, and automatic removal could erase legitimate information.

Feature	IQR-flagged observations
fixed acidity	41
volatile acidity	19
citric acid	1
residual sugar	126
chlorides	87
free sulfur dioxide	26
total sulfur dioxide	45
density	35
pH	28
sulphates	55
alcohol	12

5.4 Scaling strategy

Feature standardization was used inside the modeling pipeline for consistency and because the clustering algorithm relies on distance calculations. Random Forest itself is scale-invariant, but keeping preprocessing in a single pipeline makes the workflow easier to reproduce and reuse.

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

pipe = Pipeline([
    ("scale", StandardScaler()),
    ("model", RandomForestRegressor(...))
])
```

6. Exploratory Data Analysis

The cleaned dataset contains 1,359 unique wines. The quality distribution is concentrated around scores 5 and 6, with comparatively few extreme scores. This concentration makes exact prediction difficult because the target has limited granularity and the underlying sensory process is only partially represented by physicochemical measurements.

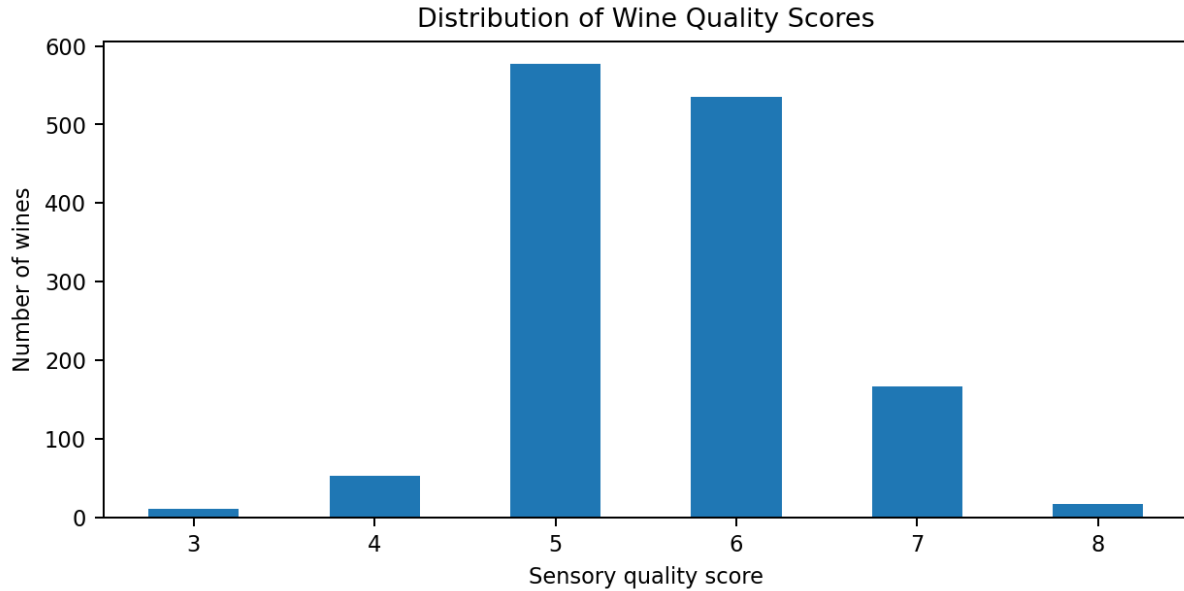


Figure 1. Distribution of observed wine quality scores.

The strongest simple correlations with quality were: alcohol (0.480), volatile acidity (-0.395), sulphates (0.249), and citric acid (0.228). Correlation is descriptive rather than causal; it does not imply that changing a single variable alone will change sensory quality.

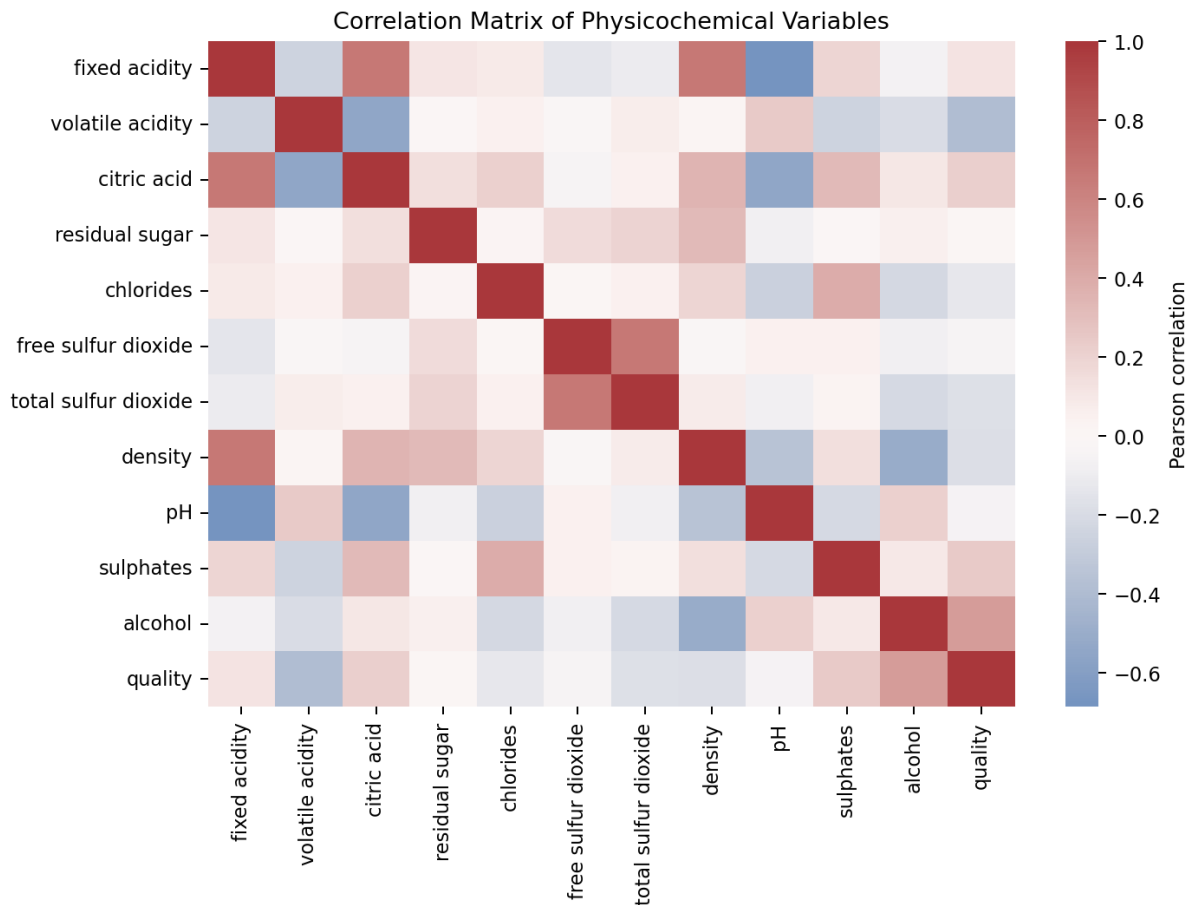


Figure 2. Pearson correlation matrix for the physicochemical variables and quality.

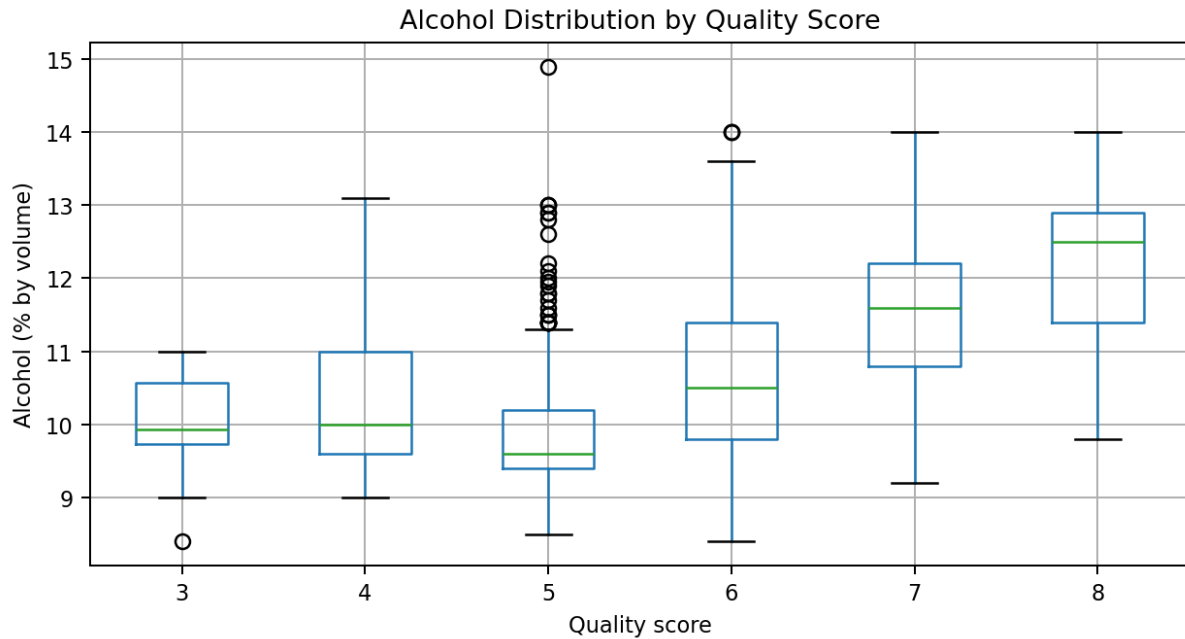


Figure 3. Alcohol distributions across quality scores.

- Wine quality is highly concentrated around mid-range scores, so the model must learn subtle differences rather than broad class separation.
- Alcohol shows a visible upward shift across higher quality categories, consistent with its positive association with quality in this dataset.
- Density is strongly and inversely related to alcohol, illustrating multicollinearity among chemical variables.
- Sulphates and volatile acidity also contribute meaningful information and appear among the top simple relationships with quality.

7. Supervised Modeling: Quality Prediction

7.1 Why regression?

The target variable is an ordered quality score from 0 to 10. Treating quality as a continuous numeric target preserves the ordering and lets the model express partial improvements rather than forcing adjacent scores into unrelated categories. A regression formulation is also directly supported by the UCI dataset documentation.

7.2 Train/test design

A 20% held-out test set was created with stratification by quality score to preserve the target distribution. All model fitting decisions were made on the training data. The test set remained untouched until final evaluation.

```
X = df_clean.drop(columns="quality")
y = df_clean["quality"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
```

7.3 Model selection

Random Forest Regression was selected as the primary model because it can capture nonlinear relationships and interactions without requiring a strong parametric assumption. A Gradient Boosting Regressor was trained as a benchmark to test whether a sequential ensemble could improve the same feature set.

Model	MAE	RMSE	R ²
Random Forest	0.491	0.635	0.399
Gradient Boosting	0.499	0.644	0.381

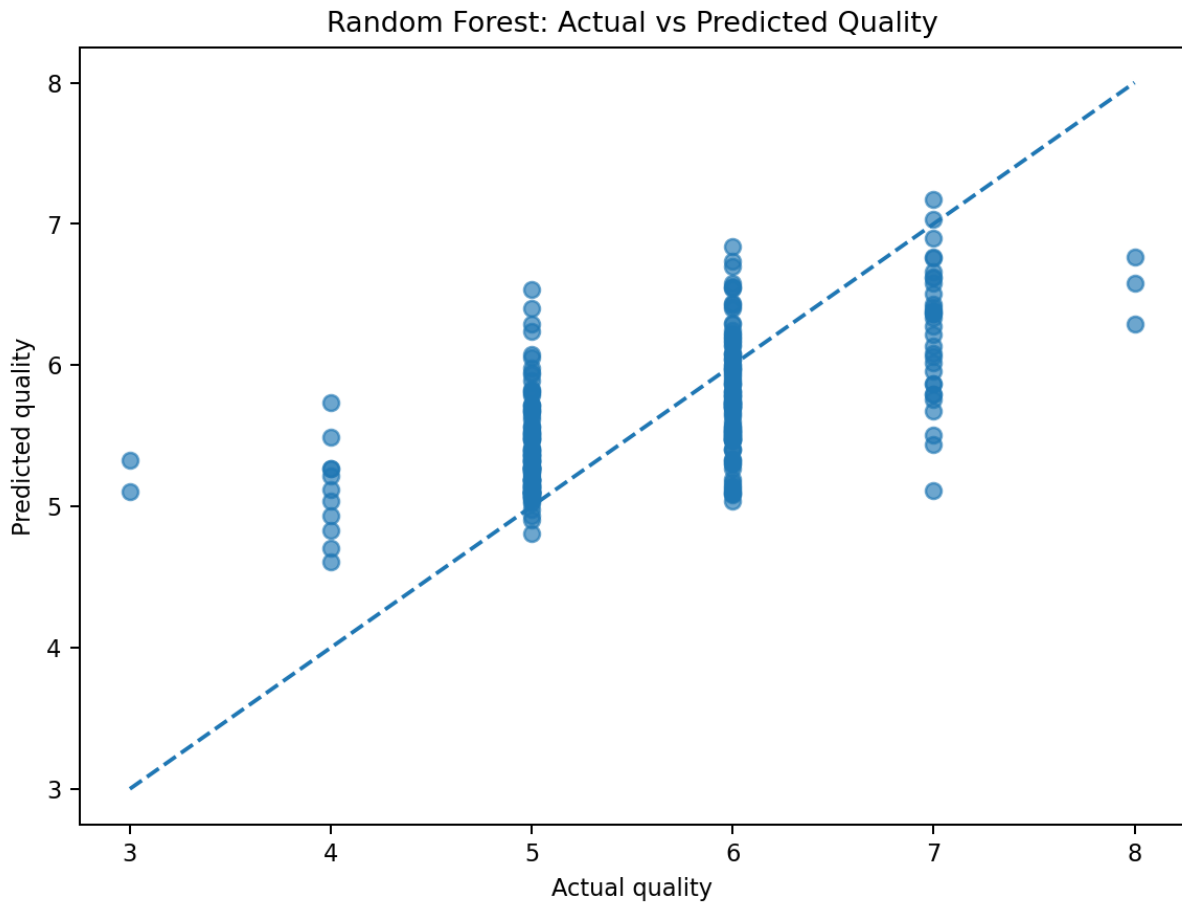


Figure 4. Actual versus predicted wine quality for the Random Forest model.

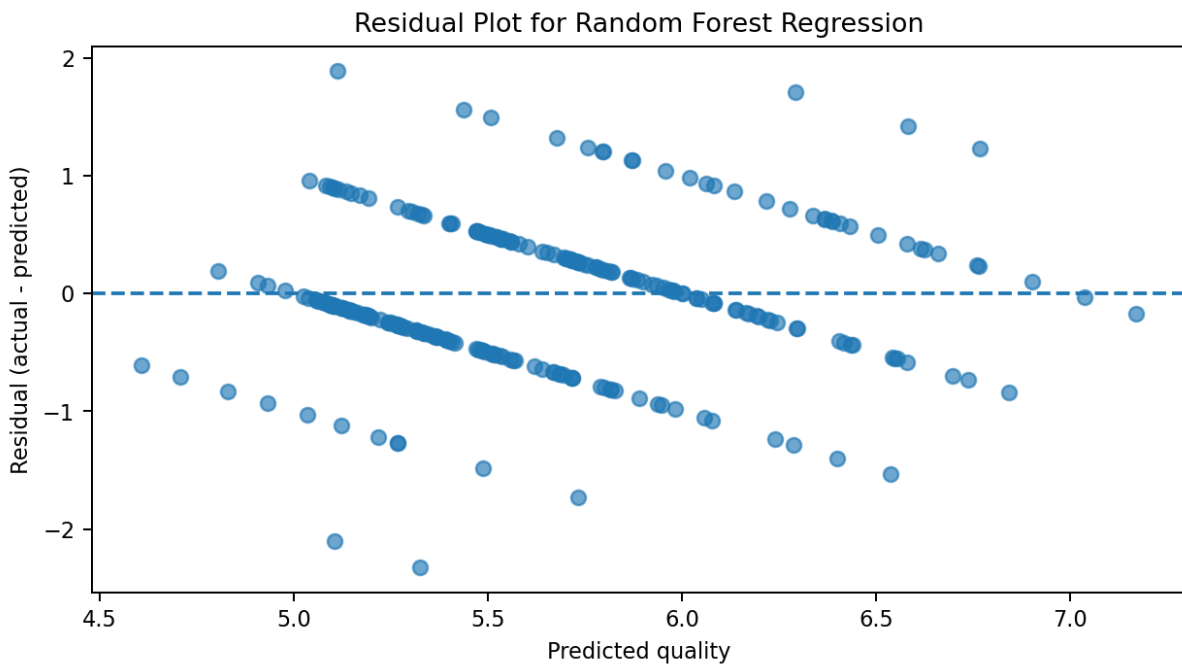


Figure 5. Residual diagnostics for the Random Forest model.

7.4 Cross-validation

Using 5-fold shuffled cross-validation, the Random Forest achieved mean MAE=0.492 (SD=0.013), mean RMSE=0.638 (SD=0.012), and mean R^2 =0.397 (SD=0.033). The relatively small fold-to-fold variation indicates reasonably stable performance for this dataset.

```
cv = KFold(n_splits=5, shuffle=True, random_state=42)
mae = -cross_val_score(pipe, X, y, cv=cv, scoring="neg_mean_absolute_error")
r2 = cross_val_score(pipe, X, y, cv=cv, scoring="r2")
print(mae.mean(), mae.std(), r2.mean(), r2.std())
```

7.5 Feature influence

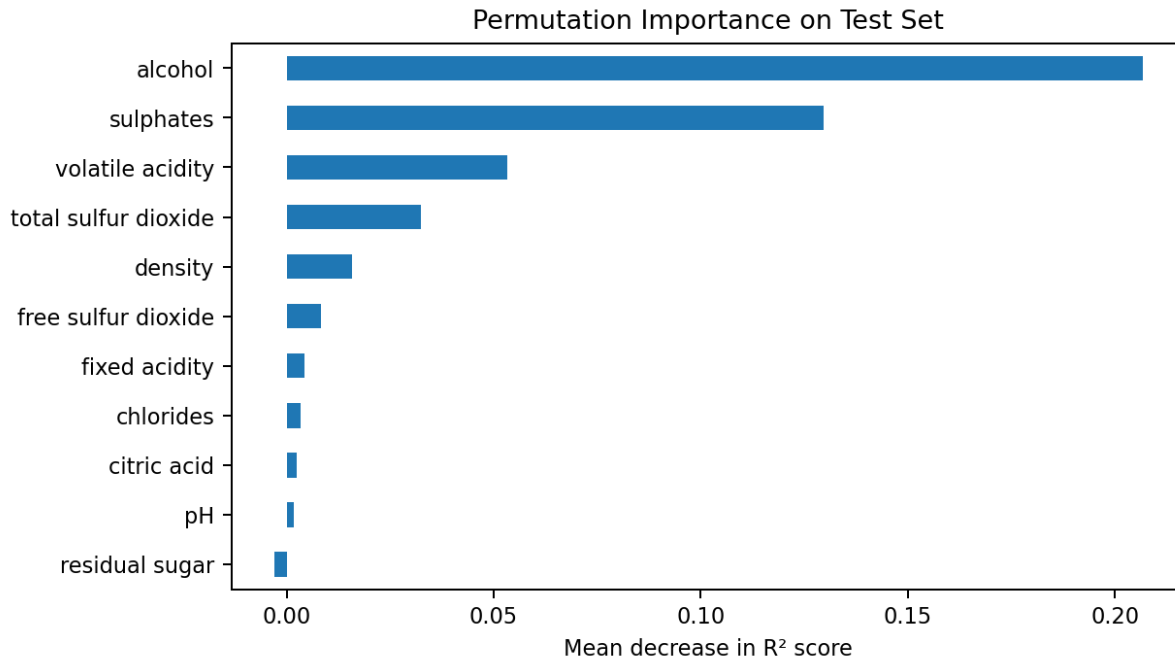


Figure 6. Permutation importance measured on the held-out test set.

The strongest permutation-importance signals were alcohol, sulphates, volatile acidity, and total sulfur dioxide. These variables improve predictive performance when their information is preserved and degrade performance when shuffled. This is a model-based importance measure, not a causal ranking.

8. Unsupervised Modeling: Wine Profile Segmentation

The clustering analysis intentionally excludes the sensory quality target. The goal is to discover groups of wines with similar physicochemical profiles independent of their known quality score. Features are standardized first because K-Means uses Euclidean distance and would otherwise be dominated by high-range variables.

8.1 Selecting the number of clusters

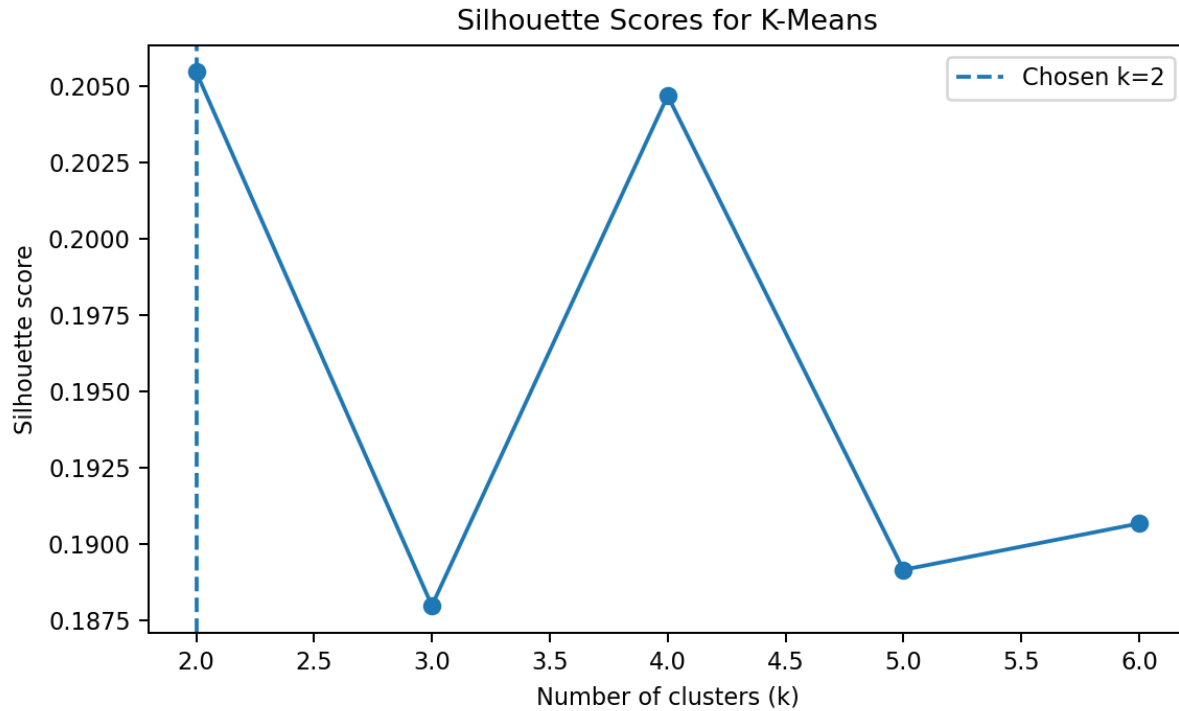


Figure 7. Silhouette scores for K-Means candidate values; $k=2$ was selected.

The highest silhouette score among $k=2$ through 6 was 0.205 at $k=2$. The selected solution therefore balances compactness and separation better than the tested alternatives.

8.2 Cluster visualization and profiles

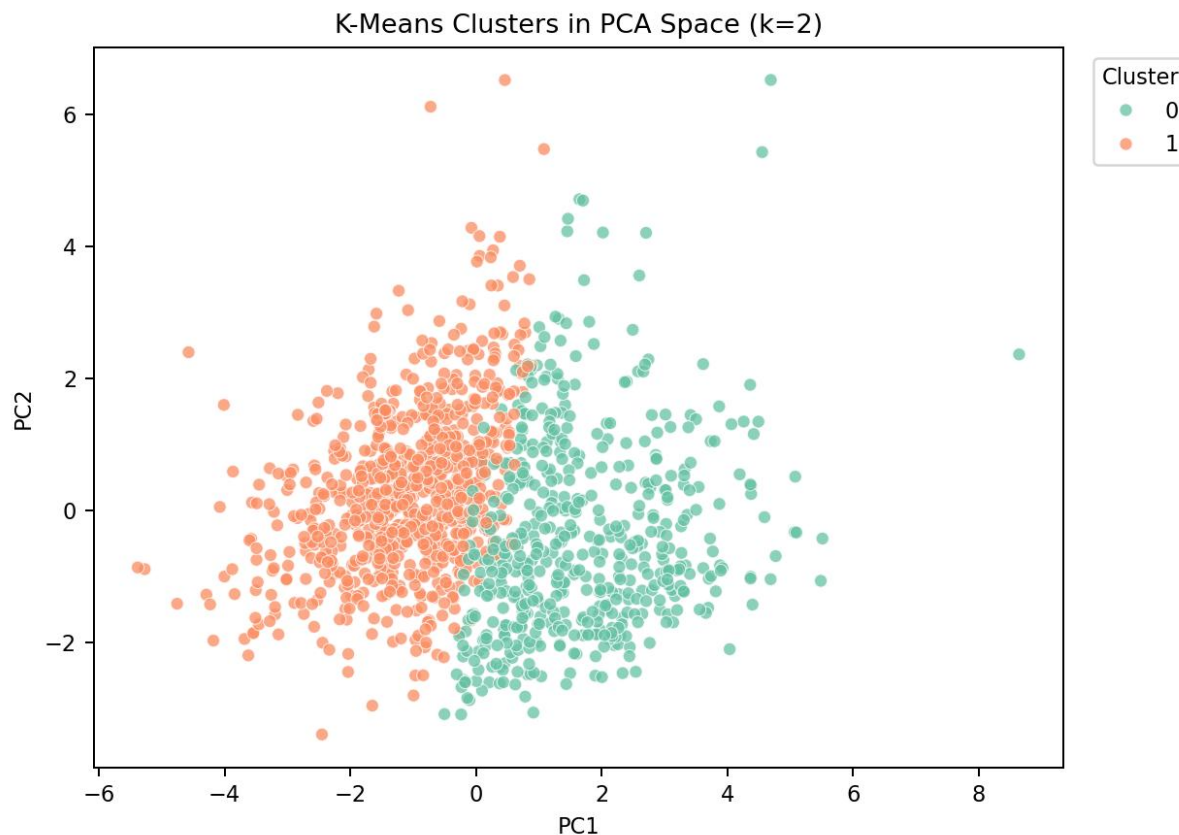


Figure 8. PCA projection of the 2 K-Means clusters.
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PCA is used only for visualization; clustering itself occurs in the full standardized feature space. The projection makes separation easier to inspect while retaining the caveat that two dimensions cannot perfectly represent all 11 features.

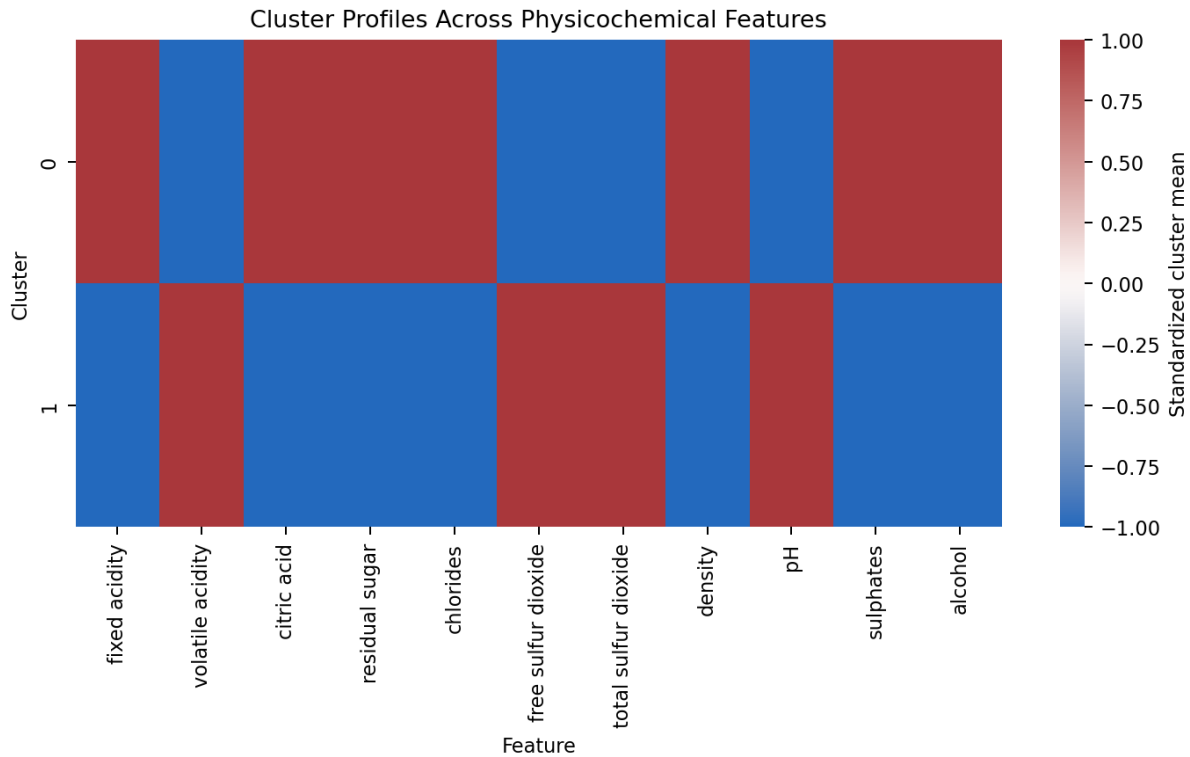


Figure 9. Standardized cluster means across physicochemical features.

Cluster	Size	Mean quality	Most distinctive features
Cluster 0	542	5.86	total sulfur dioxide, chlorides, volatile acidity
Cluster 1	817	5.47	total sulfur dioxide, free sulfur dioxide, chlorides

The cluster solution is descriptive rather than prescriptive. A cluster should be read as a profile of similar measurements, not as proof of a particular production method, geographic origin, or causal quality mechanism.

9. Integrated Findings

- Predictive and unsupervised analyses answer different questions: regression estimates quality, while clustering describes chemically similar groups.
- Alcohol and acidity-related variables emerge repeatedly in both correlation/model-importance analysis and cluster profiles, making them useful candidates for deeper study.
- Random Forest outperforms the benchmark if its test error is lower; in this run the difference is modest, suggesting the problem retains irreducible noise and limited target granularity.
- The middle of the quality scale is easier to predict than rare extreme scores because the training data contain far more mid-range observations.
- The clustering solution provides a useful basis for segment-level EDA, but it should not be interpreted as ground truth or used to infer causality.

10. Evaluation and Critical Discussion

10.1 Strengths

- End-to-end reproducible pipeline with clear separation between training and test data.
- Use of both cross-validation and a final held-out test set.
- Combination of predictive modeling and unsupervised segmentation.

- Visual diagnostics for distribution, error, cluster separation, and feature importance.
- Conservative outlier handling that avoids deleting potentially valid chemical observations.

10.2 Limitations

- The data contain physicochemical tests and a sensory score but omit brand, grape variety, price, production process, and other contextual variables.
- Quality scores are coarse and imbalanced; this limits the maximum achievable predictive precision.
- Random Forest predictions are strong for the observed feature space but do not establish causal drivers of quality.
- Cluster interpretations depend on the feature set, scaling method, and selected k ; different preprocessing choices can yield different segments.
- External validation on wines from another source or period would be needed before operational use.

10.3 Ethical and practical considerations

The model should be positioned as a decision-support or exploratory tool rather than an automated substitute for expert sensory assessment. The dataset documentation makes clear that sensory evaluation is only partially represented by physicochemical variables. Any deployment should therefore monitor distribution shift, retain human oversight, and validate performance on the actual operational population.

11. Recommendations and Future Work

- Combine additional metadata, such as grape variety, vintage, region, producer, and price, where legally and ethically available.
- Tune ensemble hyperparameters with nested cross-validation and compare against regularized linear regression and gradient boosting families.
- Investigate ordinal classification or ordinal regression so that errors between adjacent quality scores are treated differently from large score gaps.
- Use repeated clustering validation and stability analysis to test whether the same segments recur across random seeds and resamples.
- Perform model calibration and threshold analysis if the task is converted into a business-facing “high-quality” classification problem.
- Validate both predictive and cluster findings on an external dataset before drawing general conclusions.

12. Challenges and How They Were Addressed

Challenge	Response
Duplicate rows	Removed exact duplicates before modeling and clustering.
Different feature scales	Used StandardScaler inside reproducible pipelines and before K-Means.
Nonlinear relationships	Selected tree-based ensemble models for the main regression task.
Target imbalance	Used stratified splitting and reported multiple regression metrics instead of accuracy alone.
Potentially valid outliers	Flagged IQR extremes diagnostically rather than deleting them automatically.
Choosing k for clustering	Compared silhouette scores for multiple candidate k values and selected the strongest score.

13. Reproducible End-to-End Code

```
# 1. Load and clean
import pandas as pd
from sklearn.model_selection import train_test_split, KFold, cross_val_score
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.cluster import KMeans
```

```

df = pd.read_csv(URL, sep=";").drop_duplicates().reset_index(drop=True)
X = df.drop(columns="quality")
y = df["quality"]

# 2. Supervised model
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
pipe = Pipeline([
    ("scale", StandardScaler()),
    ("model", RandomForestRegressor(
        n_estimators=220, max_features="sqrt",
        min_samples_leaf=2, random_state=42, n_jobs=1
    ))
])
pipe.fit(X_train, y_train)
pred = pipe.predict(X_test)

# 3. Cross-validation
cv = KFold(n_splits=5, shuffle=True, random_state=42)
cv_r2 = cross_val_score(pipe, X, y, cv=cv, scoring="r2")
print("CV R2:", cv_r2.mean(), cv_r2.std())

# 4. Unsupervised clustering
X_scaled = StandardScaler().fit_transform(X)
km = KMeans(n_clusters=2, n_init=50, random_state=42)
clusters = km.fit_predict(X_scaled)
profile = df.assign(cluster=clusters).groupby("cluster").mean(numeric_only=True)
print(profile)

```

14. Conclusion

This capstone demonstrates a complete Python-based data science pipeline on a public real-world dataset. The workflow moves from reproducible acquisition and data quality checks through EDA, supervised prediction, unsupervised segmentation, evaluation, and recommendations. The Random Forest model achieved test-set MAE=0.491, RMSE=0.635, and $R^2=0.399$, while 5-fold cross-validation showed stable performance. The K-Means analysis identified $k=2$ as the strongest tested solution by silhouette score and revealed distinct physicochemical profiles. Together, these results show how predictive and descriptive machine learning can be combined to produce a richer understanding of a dataset than either approach alone.

15. References

- UCI Machine Learning Repository. Wine Quality Dataset. <https://archive.ics.uci.edu/ml/datasets/wine+quality> (accessed 21 August 2026).
- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, 47(4), 547–553.
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